

Lecture 18. Bayesian regression

COMP90051 Statistical Machine Learning

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This lecture

- Uncertainty not captured by point estimates
- Bayesian approach preserves uncertainty
- Sequential Bayesian updating
- Conjugate prior (Normal-Normal)
- Using posterior for Bayesian predictions on test

Training == optimisation (?)

Stages of learning & inference:

- Formulate model

Regression

$$p(y|\mathbf{x}) = \text{sigmoid}(\mathbf{x}'\mathbf{w})$$

$$p(y|\mathbf{x}) = \text{Normal}(\mathbf{x}'\mathbf{w}; \sigma^2)$$

- Fit parameters to data

$$\hat{\mathbf{w}} = \text{argmax}_{\mathbf{w}} p(\mathbf{y}|\mathbf{X}, \mathbf{w})p(\mathbf{w}) \quad \textit{ditto}$$

- Make prediction

$$p(y_*|\mathbf{x}_*) = \text{sigmoid}(\mathbf{x}'_*\hat{\mathbf{w}})$$

$$E[y_*] = \mathbf{x}'_*\hat{\mathbf{w}}$$

$\hat{\mathbf{w}}$ referred to as a '*point estimate*'

Bayesian Alternative

Nothing special about $\hat{\mathbf{w}}$... use more than one value?

- Formulate model

Regression

$$p(y|\mathbf{x}) = \text{sigmoid}(\mathbf{x}'\mathbf{w}) \quad p(y|\mathbf{x}) = \text{Normal}(\mathbf{x}'\mathbf{w}; \sigma^2)$$

- Consider the **space of likely parameters** – those that fit the training data well

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y})$$

- Make **‘expected’** prediction

$$p(y_*|\mathbf{x}_*) = E_{p(\mathbf{w}|\mathbf{X}, \mathbf{y})} [\text{sigmoid}(\mathbf{x}'_*\mathbf{w})]$$

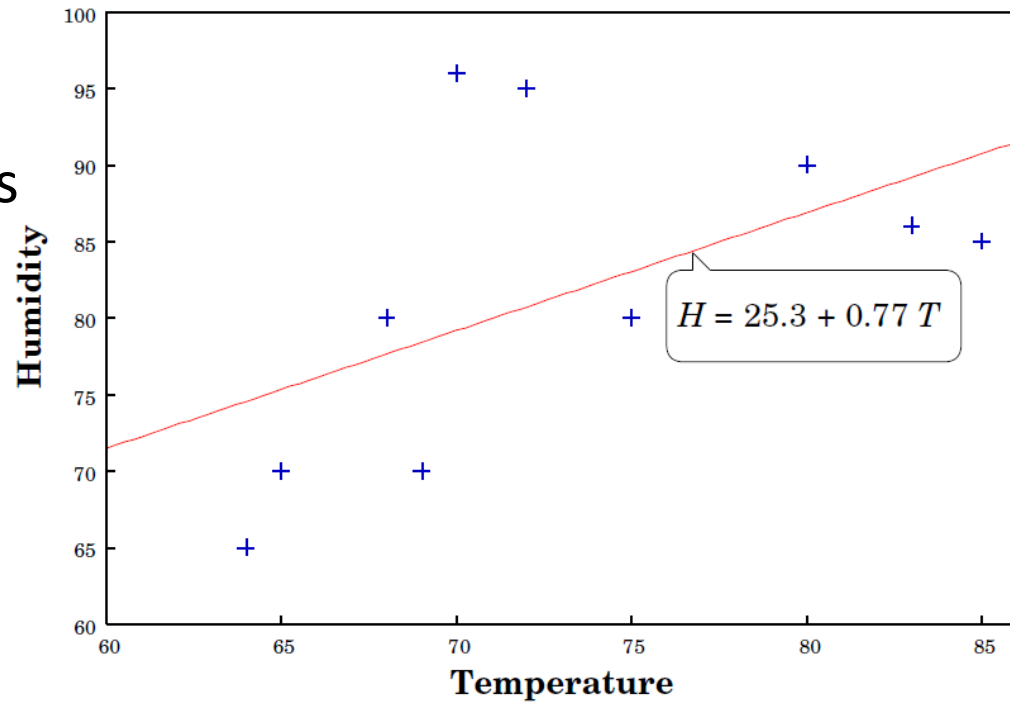
$$p(y_*|\mathbf{x}_*) = E_{p(\mathbf{w}|\mathbf{X}, \mathbf{y})} [\text{Normal}(\mathbf{x}'_*\mathbf{w}, \sigma^2)]$$

Uncertainty

From small training sets, we rarely have complete confidence in any models learned. Can we quantify the uncertainty, and use it in making predictions?

Regression Revisited

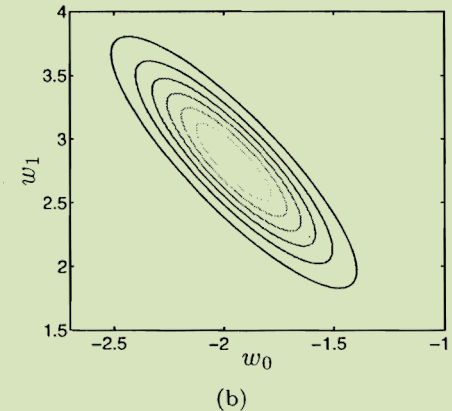
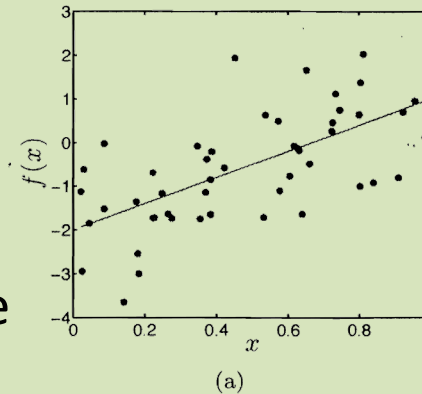
- Learn model from data
 - * minimise error residuals by choosing weights
$$\hat{\mathbf{w}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$
- But... how confident are we
 - * in $\hat{\mathbf{w}}$?
 - * in the predictions?



Linear regression: $y = w_0 + w_1 x$
(here y = humidity, x = temperature)

Do we trust point estimate $\hat{\mathbf{w}}$?

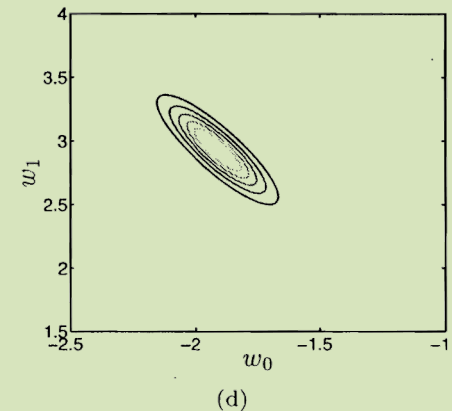
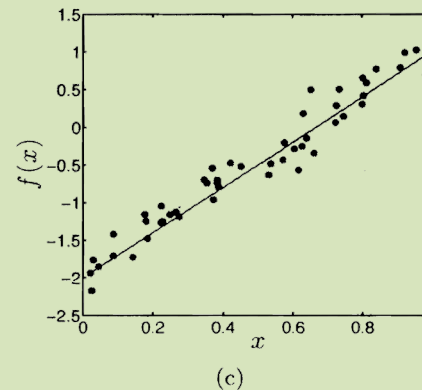
- How *stable* is learning?
 - * $\hat{\mathbf{w}}$ highly sensitive to noise
 - * how much uncertainty in parameter estimate?
 - * more *informative* if neg log likelihood objective highly peaked



- Formalised as *Fisher Information matrix*

- * $E[2^{\text{nd}} \text{ deriv of NLL}]$

$$\mathcal{I} = \frac{1}{\sigma^2} \mathbf{X}'\mathbf{X}$$



- * measures *curvature of objective* about $\hat{\mathbf{w}}$

Mini Summary

- Uncertainty not captured by point estimates (MLE, MAP)
- Uncertainty might capture range of plausible parameters
- (Frequentist) idea of Fisher information as likelihood sensitivity at point estimates

Next time: The Bayesian view (reminder)

The Bayesian View

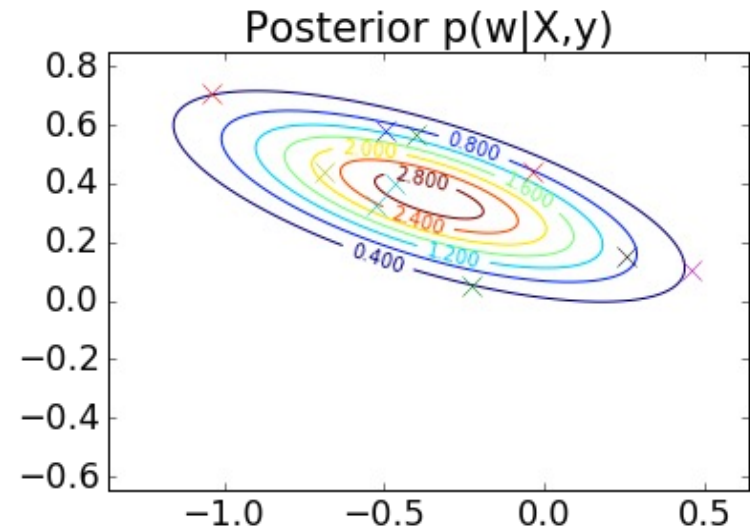
Retain and model all unknowns (e.g., uncertainty over parameters) and use this information when making inferences.

A Bayesian View

- Could we reason over *all* parameters that are consistent with the data?
 - * weights with a better fit to the training data should be more probable than others
 - * make predictions with all these weights, *scaled by their probability*
- This is the idea underlying **Bayesian** inference

Uncertainty over parameters

- Many reasonable solutions to objective
 - * why select just one?
- Reason under **all** possible parameter values
 - * weighted by their *posterior probability*
- More robust predictions
 - * less sensitive to overfitting, particularly with small training sets
 - * can give rise to more expressive model class (Bayesian logistic regression becomes non-linear!)



Frequentist vs Bayesian “divide”

- **Frequentist**: learning using *point estimates*, regularisation, *p*-values ...
 - * backed by sophisticated theory on simplifying assumptions
 - * mostly simpler algorithms, characterises much practical machine learning research
- **Bayesian**: maintain *uncertainty*, marginalise (sum) out unknowns during inference
 - * some theory
 - * often more complex algorithms, but not always
 - * often (not always) more computationally expensive

Mini Summary

- Frequentist's central preference of point estimates don't capture uncertainty
- Bayesian view is to quantify belief in prior, update it to posterior using observations

Next time: Bayesian approach to linear regression

Bayesian Regression

*Application of Bayesian inference
to linear regression, using
Normal prior over $\mathbf{w}$*

Revisiting Linear Regression

- Recall probabilistic formulation of linear regression

$\mathbf{I}_D = D \times D$ identity matrix

$$y \sim \text{Normal}(\mathbf{x}'\mathbf{w}, \sigma^2)$$

- Bayes rule:

$$\mathbf{w} \sim \text{Normal}(\mathbf{0}, \gamma^2 \mathbf{I}_D)$$

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y}) = \frac{p(\mathbf{y}|\mathbf{X}, \mathbf{w})p(\mathbf{w})}{p(\mathbf{y}|\mathbf{X})}$$

$$\max_{\mathbf{w}} p(\mathbf{w}|\mathbf{X}, \mathbf{y}) = \max_{\mathbf{w}} p(\mathbf{y}|\mathbf{X}, \mathbf{w})p(\mathbf{w})$$

- Gives rise to penalised objective (ridge regression)

point estimate taken here, avoids computing marginal likelihood term

Bayesian Linear Regression

- Rewind one step, consider full posterior

$$p(\mathbf{w} | \mathbf{X}, \mathbf{y}, \sigma^2) = \frac{p(\mathbf{y} | \mathbf{X}, \mathbf{w}, \sigma^2) p(\mathbf{w})}{p(\mathbf{y} | \mathbf{X}, \sigma^2)}$$
$$= \frac{p(\mathbf{y} | \mathbf{X}, \mathbf{w}, \sigma^2) p(\mathbf{w})}{\int p(\mathbf{y}, | \mathbf{X}, \mathbf{w}, \sigma^2) p(\mathbf{w}) d\mathbf{w}} \quad \text{☹️}$$

Here we
assume noise
var. known

- Can we compute the denominator (**marginal likelihood** or **evidence**)?
 - * if so, we can use the full posterior, not just its mode

Bayesian Linear Regression (cont)

- We have two Normal distributions
 - * normal likelihood x normal prior
- Their product is also a Normal distribution
 - * **conjugate prior**: *when product of likelihood x prior results in the same distribution as the prior*
 - * *evidence* can be computed easily using the normalising constant of the Normal distribution

$$\begin{aligned} p(\mathbf{w}|\mathbf{X}, \mathbf{y}, \sigma^2) &\propto \text{Normal}(\mathbf{w}|\mathbf{0}, \gamma^2\mathbf{I}_D)\text{Normal}(\mathbf{y}|\mathbf{X}\mathbf{w}, \sigma^2\mathbf{I}_N) \\ &\propto \text{Normal}(\mathbf{w}|\mathbf{w}_N, \mathbf{V}_N) \end{aligned}$$



closed form solution for
posterior!

Bayesian Linear Regression (cont)

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y}, \sigma^2) \propto \text{Normal}(\mathbf{w}|\mathbf{0}, \gamma^2 \mathbf{I}_D) \text{Normal}(\mathbf{y}|\mathbf{X}\mathbf{w}, \sigma^2 \mathbf{I}_N) \\ \propto \text{Normal}(\mathbf{w}|\mathbf{w}_N, \mathbf{V}_N)$$

where

$$\mathbf{w}_N = \frac{1}{\sigma^2} \mathbf{V}_N \mathbf{X}' \mathbf{y}$$

$$\mathbf{V}_N = \sigma^2 \left(\mathbf{X}' \mathbf{X} + \frac{\sigma^2}{\gamma^2} \mathbf{I}_D \right)^{-1}$$

Advanced: verify by expressing product of two Normals, gathering exponents together and 'completing the square' to express as squared exponential (i.e., Normal distribution).

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Example

- Given:** The data comes from a **Normal distribution with variance 1 but unknown mean θ** .
- Goal:** Find the **posterior over the mean** after seeing one data point where $\mathbf{X}=1$. Assume a **Normal prior over θ with mean 0 and variance 1**.

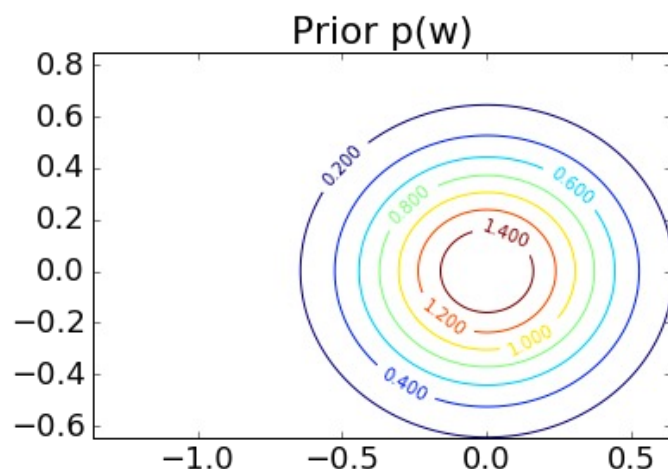
$$P(\theta|X=1) = \frac{P(X=1|\theta)P(\theta)}{P(X=1)} \\ \propto P(X=1|\theta)P(\theta) \\ = \left[\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(1-\theta)^2}{2}\right) \right] \left[\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{\theta^2}{2}\right) \right] \\ \propto \exp\left(-\frac{(1-\theta)^2 + \theta^2}{2}\right) = \exp\left(-\frac{2\theta^2 - 2\theta + 1}{2}\right) \\ = \exp\left(-\frac{\theta^2 - \theta + \frac{1}{2}}{2 \times \frac{1}{2}}\right)$$

Discard constants wrt θ

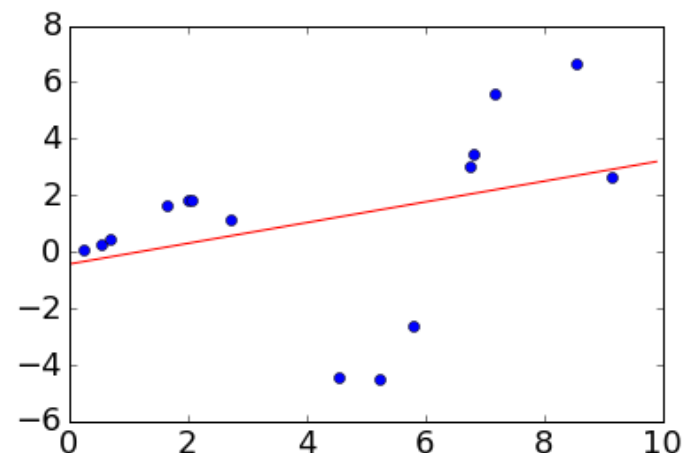
The Normal distribution:
 $\mathcal{N}(x; \mu, \sigma^2) \equiv \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

Make leading numerator zero:
 $\times \frac{1}{2}$ on top and bottom

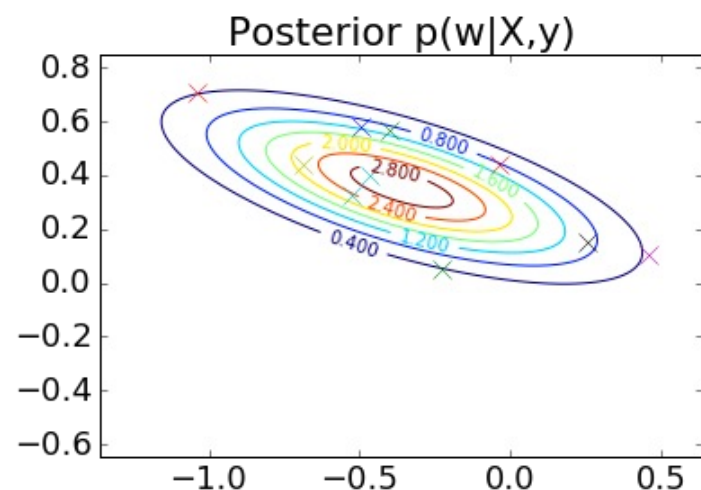
Bayesian Linear Regression example



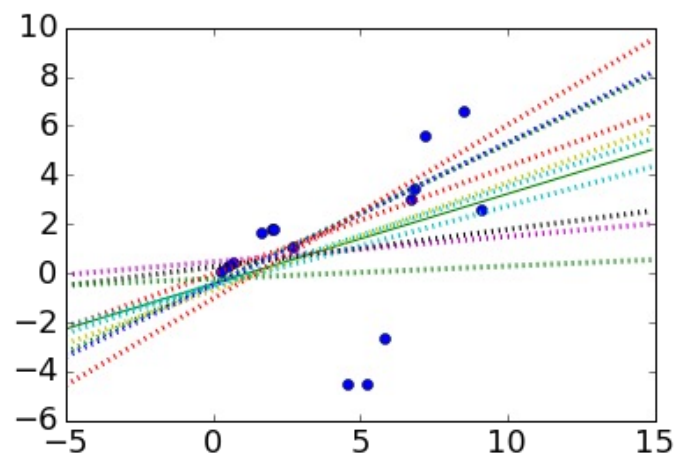
Step 1: select prior, here spherical about $\mathbf{0}$



Step 2: observe training data



Step 3: formulate posterior, from prior & likelihood

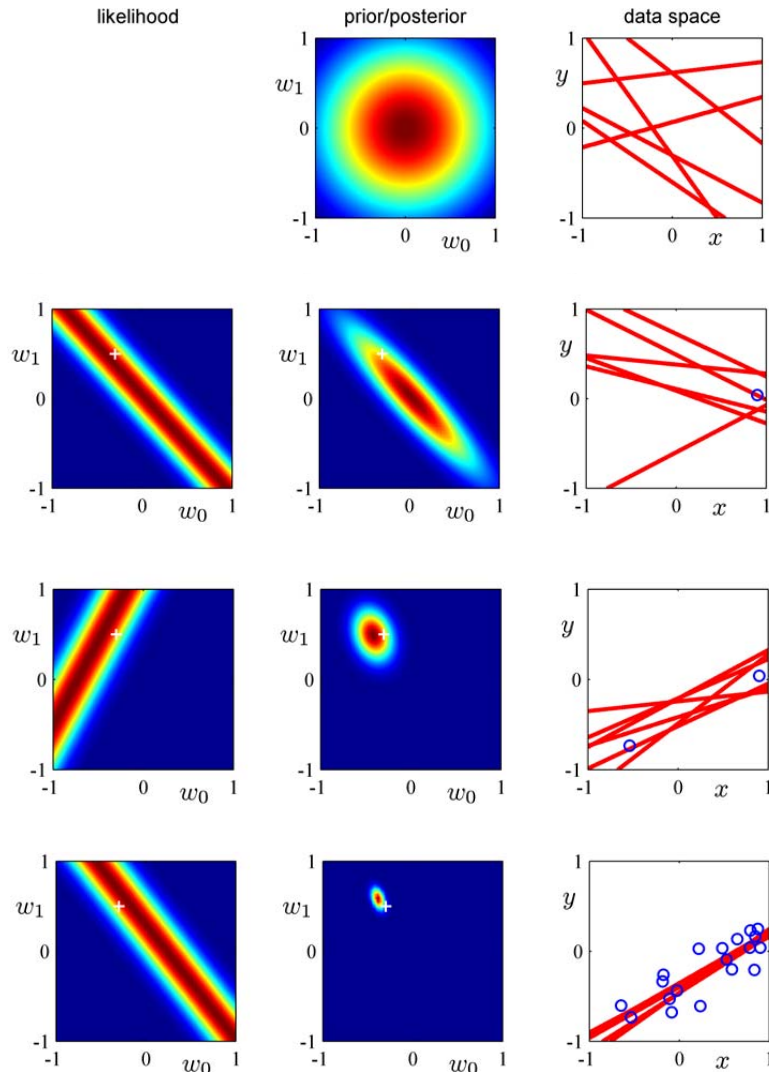


Samples from posterior

Sequential Bayesian Updating

- Can formulate $p(\mathbf{w}|\mathbf{X}, \mathbf{y}, \sigma^2)$ for given dataset
- What happens as we see more and more data?
 1. Start from prior $p(\mathbf{w})$
 2. See new labelled datapoint
 3. Compute posterior $p(\mathbf{w}|\mathbf{X}, \mathbf{y}, \sigma^2)$
 4. The *posterior now takes role of prior*
& repeat from step 2

Sequential Bayesian Updating



- Initially know little, many regression lines licensed
- Likelihood constrains possible weights such that regression is close to point
- Posterior becomes more refined/peaked as more data introduced
- Approaches a point mass

Bishop Fig 3.7, p155

Stages of Training

1. Decide on model formulation & prior
2. Compute *posterior* over parameters, $p(\mathbf{w} | \mathbf{X}, \mathbf{y})$

MAP

approx. Bayes

exact Bayes

- | | | |
|--------------------------------------|---|---|
| 3. Find <i>mode</i> for $\mathbf{w}$ | 3. Sample many $\mathbf{w}$ | 3. Use <i>all</i> $\mathbf{w}$ to make <i>expected</i> prediction on test |
| 4. Use to make prediction on test | 4. Use to make <i>ensemble</i> average prediction on test | |

Prediction with uncertain $\mathbf{w}$

- Could predict using sampled regression curves
 - * sample S parameters, $\mathbf{w}^{(s)}, s \in \{1, \dots, S\}$
 - * for each sample compute prediction $y_*^{(s)}$ at test point $\mathbf{x}_*$
 - * compute the mean (and var.) over these predictions
 - * this process is known as **Monte Carlo integration**
- For Bayesian regression there's a simpler solution
 - * integration can be done analytically, for

$$p(\hat{y}_* | \mathbf{X}, \mathbf{y}, \mathbf{x}_*, \sigma^2) = \int p(\mathbf{w} | \mathbf{X}, \mathbf{y}, \sigma^2) p(y_* | \mathbf{x}_*, \mathbf{w}, \sigma^2) d\mathbf{w}$$

Prediction (cont.)

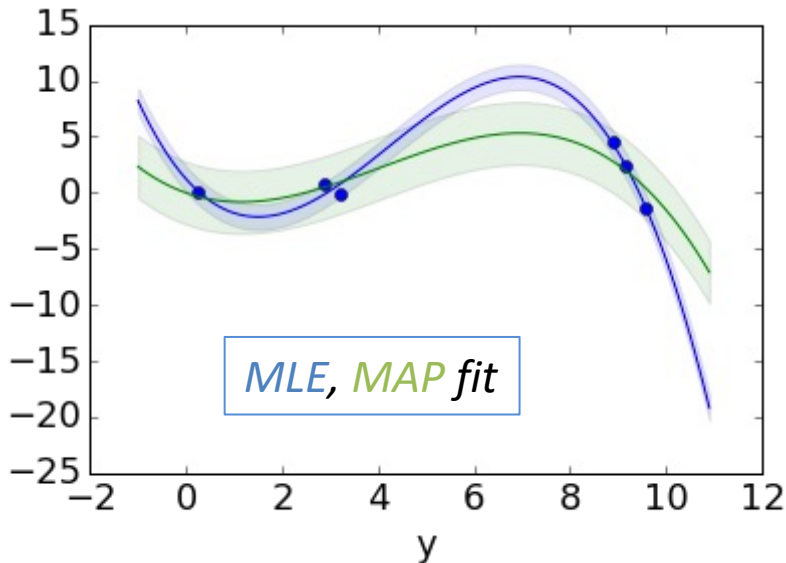
- Pleasant properties of Gaussian distribution means integration is tractable

$$\begin{aligned} p(y_* | \mathbf{x}_*, \mathbf{X}, \mathbf{y}, \sigma^2) &= \int p(\mathbf{w} | \mathbf{X}, \mathbf{y}, \sigma^2) p(y_* | \mathbf{x}_*, \mathbf{w}, \sigma^2) d\mathbf{w} \\ &= \int \text{Normal}(\mathbf{w} | \mathbf{w}_N, \mathbf{V}_N) \text{Normal}(y_* | \mathbf{x}'_* \mathbf{w}, \sigma^2) d\mathbf{w} \\ &= \text{Normal}(y_* | \mathbf{x}'_* \mathbf{w}_N, \sigma_N^2(\mathbf{x}_*)) \\ \sigma_N^2(\mathbf{x}_*) &= \sigma^2 + \mathbf{x}'_* \mathbf{V}_N \mathbf{x}_* \end{aligned}$$

- * additive variance based on $\mathbf{x}_*$ match to training data
- * **cf. MLE/MAP estimate, where variance is a fixed constant**

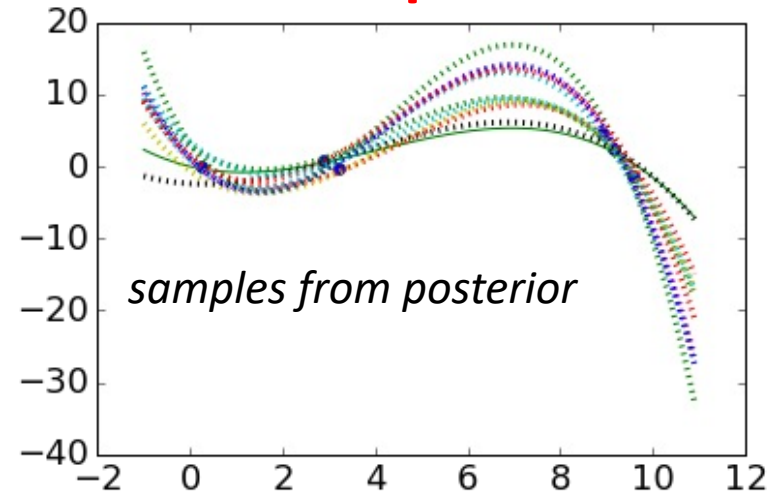
Bayesian Prediction example

Point estimate

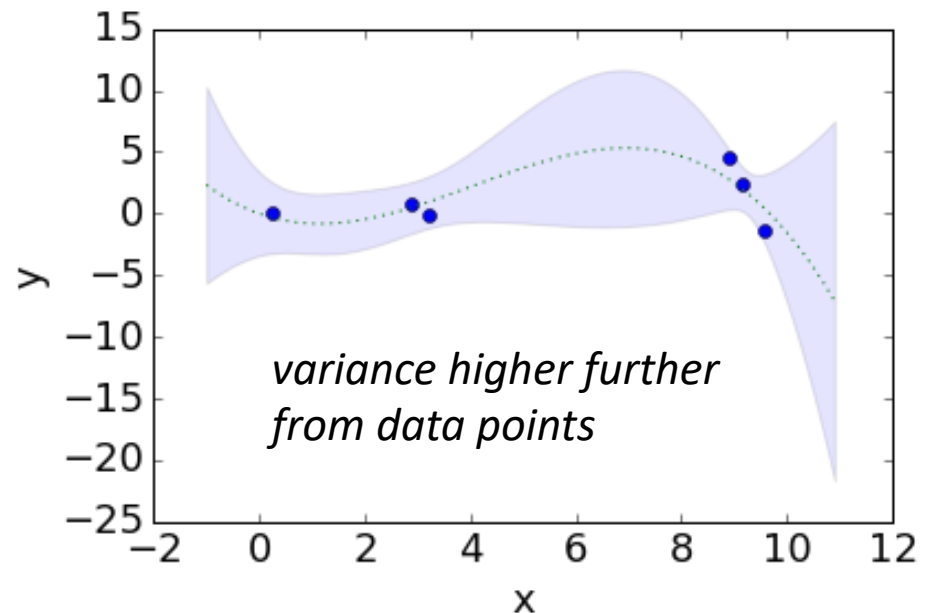


MLE (blue) and MAP (green)
point estimates, *with fixed
variance*

Data: $y = x \sin(x)$; Model = cubic



Bayesian inference



Caveats

- Assumptions
 - * known data noise parameter, σ^2
 - * data was drawn from the model distribution
- In real settings, σ^2 is unknown
 - * has its own conjugate prior
Normal likelihood $\times$ *InverseGamma* prior
results in *InverseGamma* posterior
 - * closed form predictive distribution, with student-T likelihood
(see *Murphy*, 7.6.3)

Mini Summary

- Uncertainty not captured by point estimates (MLE, MAP)
- Bayesian approach preserves uncertainty
 - * care about predictions NOT parameters
 - * choose prior over parameters, then model posterior
- New concepts:
 - * sequential Bayesian updating
 - * conjugate prior (Normal-Normal)
- Using posterior for Bayesian predictions on test

Next time: Bayesian classification, then PGMs